

TCS3351 Cryptography and Data Security
Tutorial 9
Modern Asymmetric Ciphers

1. Given $a = [15\ 25\ 50\ 100\ 200\ 400\ 800]$ and $s = 450$. Find tuple x using inv-knapsackSum algorithm.
2. Given the super-increasing tuple $b = [7\ 11\ 23\ 43\ 87\ 173\ 357]$, $r = 41$, and modulus $n = 1001$, encrypt and decrypt the letter 'a' using the knapsack cryptosystem. Use $[7\ 6\ 5\ 1\ 2\ 3\ 4]$ as the permutation table.
3. Perform key generation, encryption and decryption using RSA algorithm for the following problem.

$$p = 11; q = 13; e = 7; \text{Message} = FCI$$

4. Perform key generation, encryption and decryption using RSA algorithm for the following problem.

$$p = 7; q = 11; e = 17; M = 8$$

5. Suppose Alice publishes her RSA public keys $(e, n) = (103, 2038667)$.
 - a) Bob wishes to send Alice the message $m = 892383$. What ciphertext does Bob send to Alice?
 - b) Alice knows that her modulus $n = 2038667$ has a prime factor of $p = 1301$. Compute the decrypt key d for Alice.
 - c) Alice receives the ciphertext $c = 317730$ from another sender Charlie. Recover the message for Alice.

References:

1. Forouzan, B. A. & Mukhopadhyay, D. (2015). Cryptography & Network Security (3rd ed.). McGraw-Hill.
2. William Stallings, "Cryptography and Network Security – Principles and Practice", Seventh Edition, Prentice Hall, 2017